

Notes on Gauge Theory

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5 Transgressions and the Chern-Simons Form

Notation: ω, Ω are forms on the bundle P , while A, F are forms on M .

5.1 Gauss-Bonnet Theorem

Our story begins with the Gauss-Bonnet theorem, which is a great theorem connecting local geometric quantity with global topologic quantity.

Theorem 5.1 (Gauss-Bonnet Theorem). *Let M^2 be a closed oriented surface with Riemannian metric, K be its Gauss curvature. Then*

$$\frac{1}{2\pi} \int_M K d\text{Vol}_M = \chi(M),$$

where $\chi(M)$ is the Euler characteristic of M . If we denote the curvature 2-form of the Levi-Civita connection by F , then $F = K d\text{Vol}_M$, thus we can rewrite the Gauss-Bonnet theorem as

$$\frac{1}{2\pi} \int_M F = \chi(M).$$

To prove this theorem, we need to use the vector field V on M .

Definition 5.2. Let V be a smooth vector field on M with isolated zeros $\{x_i\}$. On $M - \{x_i\}$, we can normalize V to be a unit vector field $\tilde{V} = \frac{V}{|V|}$. On each zero x_i , choose a small neighborhood $B_r(x_i)$, then $\tilde{V}|_{\partial B_r(x_i)}$ can be viewed as a map from S^1 to S^1 . We define the index of V at x_i as the degree of this map, i.e.

$$\text{Ind}_{x_i}(V) = \deg(\tilde{V}|_{\partial B_r(x_i)} : S^1 \rightarrow S^1).$$

The index of V is defined to be the sum of the indices at all zeros,

$$\text{Ind}(V) = \sum_i \text{Ind}_{x_i}(V).$$

The following picture shows some zeros with different indices.

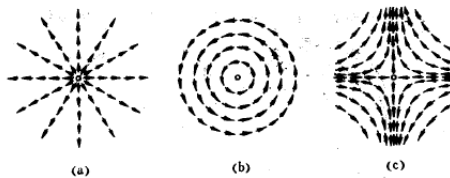


图 89

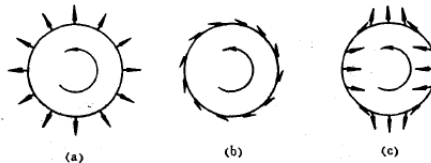


图 90

Remark. The index of V at x_i is independent of the choice of $B_r(x_i)$, thus is well-defined. Since $\text{Ind}(V)$ is an integer, it is invariant under homotopy of V .

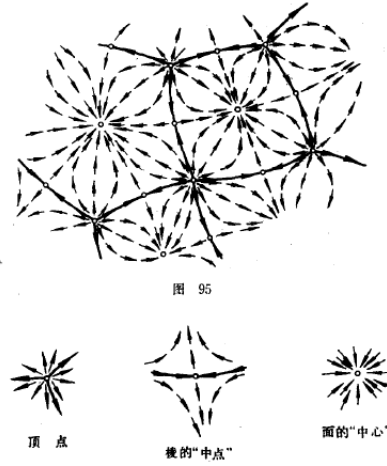
Theorem 5.3. (*Poincaré-Hopf Index Theorem*) Let M be a closed oriented surface, V be a smooth vector field on M with isolated zeros. Then

$$\text{Ind}(V) = \chi(M).$$

Sketch of the proof.

Step 1: Independence of V . Omit.

Step 2: Construction of a special V . Given a triangulation of M , we can construct a vector field V such that $\text{Ind}(V) = \chi(M)$, thus we have $\text{Ind}(V) = \chi(M)$ for any V . The following picture shows how to construct such a vector field V : $\square$



Proof of Gauss-Bonnet Theorem. Let V be a smooth vector field on M with isolated zeros $\{x_i\}$. We construct another smooth vector field W with the same zeros such that $\{\tilde{V}, \tilde{W}\} := \{\frac{V}{|V|}, \frac{W}{|W|}\}$ is an orthonormal frame on $M - \{x_i\}$ compatible with the orientation of M .

Let A be the connection 1-form of the Levi-Civita connection with respect to the frame $\{V, W\}$, i.e.,

$$A(X) := \langle \nabla_X \tilde{V}, \tilde{W} \rangle, \quad \forall X \in T(M - \{x_i\}).$$

then the curvature 2-form $F = dA$. Let $B_r(x_i)$ be a small neighborhood of x_i containing no other zeros, then

$$\begin{aligned} \frac{1}{2\pi} \int_M F &= \lim_{r \rightarrow 0} \frac{1}{2\pi} \int_{M - \cup_i B_r(x_i)} F \\ &= \lim_{r \rightarrow 0} \frac{1}{2\pi} \int_{M - \cup_i B_r(x_i)} dA \\ &= \lim_{r \rightarrow 0} \sum_i \frac{1}{2\pi} \int_{\partial(M - B_r(x_i))} A \\ &= \lim_{r \rightarrow 0} \sum_i \frac{1}{2\pi} \int_{\partial B_r(x_i)} A \end{aligned}$$

Notice that the left hand side is independent of the choice of metric and connection (by characteristic class theory), thus we can choose any metric and connection we like. In particular, we can choose the Euclidean metric and the trivial connection in each neighborhood $B_r(x_i)$. Under a local coordinate system $(B_r(x_i); x, y)$, we can write

$$\tilde{V} = \cos \theta \frac{\partial}{\partial x} + \sin \theta \frac{\partial}{\partial y}, \quad \tilde{W} = -\sin \theta \frac{\partial}{\partial x} + \cos \theta \frac{\partial}{\partial y},$$

where $\theta = \theta(x, y)$ is a smooth function on $B_r(x_i)$. Then

$$\begin{aligned} A\left(\frac{\partial}{\partial x}\right) &= \langle \nabla_{\frac{\partial}{\partial x}} \tilde{V}, \tilde{W} \rangle = \langle -\theta_x \sin \theta \frac{\partial}{\partial x} + \theta_x \cos \theta \frac{\partial}{\partial y}, -\sin \theta \frac{\partial}{\partial x} + \cos \theta \frac{\partial}{\partial y} \rangle = \theta_x, \\ A\left(\frac{\partial}{\partial y}\right) &= \langle \nabla_{\frac{\partial}{\partial y}} \tilde{V}, \tilde{W} \rangle = \langle -\theta_y \sin \theta \frac{\partial}{\partial x} + \theta_y \cos \theta \frac{\partial}{\partial y}, -\sin \theta \frac{\partial}{\partial x} + \cos \theta \frac{\partial}{\partial y} \rangle = \theta_y. \end{aligned}$$

Thus we have $A = \theta_x dx + \theta_y dy = d\theta$, so

$$\lim_{r \rightarrow 0} \frac{1}{2\pi} \int_{\partial B_r(x_i)} A = \lim_{r \rightarrow 0} \frac{1}{2\pi} \int_{\partial B_r(x_i)} d\theta = \text{Ind}_{x_i}(V).$$

Therefore,

$$\frac{1}{2\pi} \int_M F = \sum_i \text{Ind}_{x_i}(V) = \text{Ind}(V) = \chi(M). \quad \square$$

Remark. A is not a globally defined 1-form on M , but its exterior derivative dA can be extended to a globally defined 2-form on M .

Gauss-Bonnet-Chern Theorem

The Gauss-Bonnet-Chern theorem is a generalization of the Gauss-Bonnet theorem to higher dimensions.

Theorem 5.4 (Gauss-Bonnet-Chern Theorem). *Let M^{2n} be a closed oriented Riemannian manifold, F be the curvature 2-form of the Levi-Civita connection. Then*

$$\int_M \text{Pf}\left(\frac{F}{2\pi}\right) = \chi(M),$$

where Pf is the Pfaffian of a skew-symmetric matrix.

To prove this theorem, we need to use the following generalization of the Poincaré-Hopf index theorem.

Theorem 5.5 (General Poincaré-Hopf Index Theorem). *Let M be a closed oriented manifold, V be a smooth vector field on M with isolated zeros. Then*

$$\text{Ind}(V) = \chi(M).$$

Sketch of the proof.

Step 1: Independence of V . Omit.

Step 2: Construction of a special V . Choose V to be the gradient vector field of a Morse function f on M , then the zeros of V are exactly the critical points of f , and the index of V at a critical point x_i is $(-1)^{\lambda_i}$, where λ_i is the Morse index of x_i . So

$$\text{Ind}(V) = \sum_i (-1)^{\lambda_i} = \sum_{\lambda} (-1)^{\lambda} \#\{\text{critical points of index } \lambda\} = \chi(M). \quad \square$$

Sketch of the proof of G.B.C. Theorem. Consider the unit sphere bundle $\pi_S : SM \rightarrow M$, Chern constructed a global angular form Ψ on SM such that:

- Ψ is the "transgression" of $\text{Pf}\left(\frac{F}{2\pi}\right)$, i.e.,

$$d\Psi = \pi_S^* \text{Pf}\left(\frac{F}{2\pi}\right). \quad (5.1)$$

- For any $x \in M$,

$$\int_{S_x^{2n-1}} \iota_x^* \Psi = 1, \quad (5.2)$$

where $\iota_x : S_x^{2n-1} \hookrightarrow SM$ is the inclusion of the fiber at x .

Now let V be a normal vector field on M with isolated singular points $\{x_i\}$. V can be viewed as a section of SM over $M - \{x_i\}$. Let $B_r(x_i)$ be a small neighborhood of x_i containing no other singular points, then

$$\begin{aligned} \int_M \text{Pf}\left(\frac{F}{2\pi}\right) &= \lim_{r \rightarrow 0} \int_{M - \cup_i B_r(x_i)} \text{Pf}\left(\frac{F}{2\pi}\right) \\ &= \lim_{r \rightarrow 0} \int_{M - \cup_i B_r(x_i)} V^* \pi_S^* \text{Pf}\left(\frac{F}{2\pi}\right) \\ &= \lim_{r \rightarrow 0} \int_{M - \cup_i B_r(x_i)} V^* d\Psi \\ &= \lim_{r \rightarrow 0} \sum_i \int_{\partial(M - B_r(x_i))} V^* \Psi \\ &= \lim_{r \rightarrow 0} \sum_i \int_{\partial B_r(x_i)} V^* \Psi \end{aligned}$$

Notice that the last integral actually computes the degree of $V|_{\partial B_r(x_i)}$, where

$$V|_{\partial B_r(x_i)} : \partial B_r(x_i) \rightarrow SM|_{B_r(x_i)} \cong B_r(x_i) \times S^{2n-1} \simeq S^{2n-1},$$

Therefore,

$$\int_M \text{Pf}\left(\frac{F}{2\pi}\right) = \sum_i \deg(V|_{\partial B_r(x_i)}) = \sum_i \text{Ind}_{x_i}(V) = \text{Ind}(V) = \chi(M). \quad \square$$

This proof highly depends on the magical construction of the global angular form Ψ , which is a special case of the transgression form we'll talk about later.

Remark. Back in the 2-dimensional case, the unit sphere bundle SM is just the oriented orthonormal frame bundle $SO(TM)$, both of which are S^1 -bundles over M . Let ω be the Ehresmann connection of $SO(TM)$ induced by the Levi-Civita connection, Ω be the curvature of ω . Since $SO(2)$ is abelian, we have

$$\Omega = d\omega + \frac{1}{2}[\omega, \omega] = d\omega, \quad \text{Pf}\left(\frac{\Omega}{2\pi}\right) = d \text{Pf}\left(\frac{\omega}{2\pi}\right) =: d\Psi.$$

The relation between $\Omega \in \Omega^2(SO(TM), \mathfrak{so}(2))$ and $F \in \Omega^2(M)$ is

$$\text{Pf}\left(\frac{\Omega}{2\pi}\right) = \pi_S^*\left(\frac{F}{2\pi}\right).$$

So

$$d\Psi = \pi_S^*\left(\frac{F}{2\pi}\right). \quad (5.3)$$

Notice that the inclusion $\iota_x : S_x^1 \hookrightarrow SO(TM)$ pulls back ω to the Maurer-Cartan form ω_{MC} on S^1 , where

$$\begin{aligned} \omega_{MC} &= g^{-1}dg = \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix} d \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \\ &= \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} -\sin \theta d\theta & -\cos \theta d\theta \\ \cos \theta d\theta & -\sin \theta d\theta \end{pmatrix} = \begin{pmatrix} 0 & d\theta \\ -d\theta & 0 \end{pmatrix} \end{aligned}$$

Thus

$$\iota_x^* \Psi = \iota_x^* \text{Pf}\left(\frac{\omega}{2\pi}\right) = \text{Pf}\left(\frac{\iota_x^* \omega}{2\pi}\right) = \text{Pf}\left(\frac{\omega_{MC}}{2\pi}\right) = \frac{d\theta}{2\pi}, \quad (5.4)$$

is the angular form on S_x^1 with integral 1. Formulas (5.3) and (5.4) are exactly the properties of the global angular form Ψ in the proof of the G.B.C. theorem. The form A is just the pullback of Ψ via the section $V : M \rightarrow \{x_i\} \rightarrow SO(TM)$, i.e., $A = V^* \Psi$.

5.2 Transgressions

Now let's focus on a G -principal bundle $\pi : P \rightarrow M$ with connection A and curvature F . The Chern-Weil theory tells us any invariant polynomial Φ on $\text{Inv}^k(\mathfrak{g})$ defines a closed $2k$ -form $\Phi(F)$ on M and its cohomology class $[\Phi(F)]$ is independent of the choice of connection.

Consider the pullback bundle π^*P over P ,

$$\begin{array}{ccc} \pi^*P & & P \\ \downarrow & & \downarrow \pi \\ P & \xrightarrow{\pi} & M \end{array}$$

π^*P can be identified with the fiber product $P \times_M P$, thus has a tautological section $\Delta : P \rightarrow \pi^*P$, $\Delta(p) = (p, p)$. This means π^*P is a trivial bundle, so all characteristic classes of π^*P vanish. In particular, $[\Phi(\pi^*F)] = 0$ in $H^{2k}(P)$. Thus there exists a $(2k-1)$ -form $\Psi(A, F)$ on P such that

$$d\Psi(A, F) = \Phi(\pi^*F) = \pi^*\Phi(F).$$

The form $\Psi(A, F)$ is called the transgression of $\Phi(F)$.

Remark. Let $SO(TM)$ be the orthonormal frame bundle of a Riemannian manifold M^{2n} , then the transgression of $\text{Pf}\left(\frac{F}{2\pi}\right) \in \Omega^{2n}(M)$ is a $(2n-1)$ -form Ψ on $SO(TM)$ such that $d\Psi = \pi^*\text{Pf}\left(\frac{F}{2\pi}\right)$. The unit sphere bundle SM is the associated bundle of $SO(TM)$ with $SO(2n)$ naturally acting on S^{2n-1} , thus we can induce a form $\tilde{\Psi}$ on SM from Ψ . It turns out that $\tilde{\Psi}$ is exactly the global angular form Chern constructed in the proof.

There is a more explicit way to construct the transgression $\Psi(A, F)$. To introduce it, we need to review the Chern-Weil theory.

Given two connection ∇_0 and ∇_1 on vector bundle E , the independence of characteristic classes on the choice of connection is proved by the following formula:

$$\text{Tr}(\Omega_1^k) - \text{Tr}(\Omega_0^k) = d \int_0^1 k \text{Tr} \left(\frac{d\omega_t}{dt} \wedge \Omega_t^{k-1} \right) dt.$$

Here $\omega_t = (1-t)\omega_0 + t\omega_1$ is the connection form of $\nabla_t = (1-t)\nabla_0 + t\nabla_1$, Ω_t is the curvature of ∇_t . If we fix ∇_0 to be flat, i.e., $\Omega_0 = 0$, then the right hand side exactly gives a transgression of $\text{Tr}(\Omega_1^k)$.

5.3 Chern-Simons Form

Historically, Simons wanted to study the signature of a 4-manifold X^4 , which is related to the first Pontryagin class p_1 because of **the Hirzebruch signature theorem**:

$$\sigma(X) = \frac{1}{3} \int_X p_1(TX).$$

Notice that

$$p_1(TX) = f_2\left(\frac{F}{2\pi}\right) = \frac{1}{2} \left[\left(\text{Tr} \left(\frac{F}{2\pi} \right) \right)^2 - \text{Tr} \left(\frac{F^2}{4\pi^2} \right) \right] = -\frac{1}{8\pi^2} \text{Tr}(F^2).$$

Simons constructed a transgression of $p_1(TX)$, which is now called the Chern-Simons form.

Let M be a closed oriented 3-manifold, there is a fact that its oriented orthonormal frame bundle $SO(TM)$ is always trivial. So we can choose a global section $s : M \rightarrow SO(TM)$, and fix a trivial connection ∇_0 on $SO(TM)$ such that its connection form is zero with respect to s . For any connection ∇_1 , let A be its connection form with respect to s . Now the connection $\nabla_t = (1-t)\nabla_0 + t\nabla_1$ has connection form

$$A_t = tA,$$

and curvature

$$F_t = dA_t + \frac{1}{2}[A_t, A_t] = t dA + t^2 A \wedge A.$$

So the Chern-Simons form is

$$\begin{aligned} -\frac{1}{8\pi^2} \int_0^1 k \operatorname{Tr} \left(\frac{dA_t}{dt} \wedge F_t^{k-1} \right) dt &= -\frac{1}{8\pi^2} \int_0^1 2 \operatorname{Tr} (A \wedge (t dA + t^2 A \wedge A)) dt \\ &= -\frac{1}{8\pi^2} \operatorname{Tr} \left(A \wedge dA + \frac{2}{3} A \wedge A \wedge A \right). \end{aligned} \quad (5.5)$$

Remark. The Chern-Simons form is a 3-form on the oriented 3-manifold M . It is closed for the dimension reason.

Definition 5.6 (Chern-Simons functional). Let M be a closed oriented 3-manifold, $SO(TM)$ be its oriented orthonormal frame bundle, ∇ be a connection on $SO(TM)$. For any global trivialization $s : M \rightarrow SO(TM)$, let A be the connection form of ∇ with respect to s , then the Chern-Simons functional of ∇ is defined to be

$$CS(A) := \frac{1}{8\pi^2} \int_M \operatorname{Tr} \left(A \wedge dA + \frac{2}{3} A \wedge A \wedge A \right).$$

Remark. Notice that different choices of global trivialization s, s' differ by a gauge transformation $g : M \rightarrow SO(3)$. One can show that although $CS(A) \neq CS(A')$ in general, but $CS(A) - CS(A')$ is always an integer. Thus $CS(A)$ is well-defined as an element in $\mathbb{R}/\mathbb{Z}$.

Critical points of the Chern-Simons functional

If the connection ∇ is a critical point of the Chern-Simons functional. Fix a global trivialization $s : M \rightarrow SO(TM)$ and let A be the connection form of ∇ w.r.t. s . Then for any variation $a \in \Omega^1(M, \mathfrak{so}(3))$, the perturbation $\nabla + ta$ has connection form $A + ta$, so

$$\begin{aligned} &\left. \frac{d}{dt} \right|_{t=0} CS(A + ta) \\ &= \left. \frac{d}{dt} \right|_{t=0} \frac{1}{8\pi^2} \int_M \operatorname{Tr} \left((A + ta) \wedge d(A + ta) + \frac{2}{3} (A + ta) \wedge (A + ta) \wedge (A + ta) \right) \\ &= \frac{1}{8\pi^2} \int_M \operatorname{Tr} \left(a \wedge dA + A \wedge da + \frac{2}{3} (a \wedge A \wedge A + A \wedge a \wedge A + A \wedge A \wedge a) \right) \\ &= \frac{1}{8\pi^2} \int_M \operatorname{Tr} (a \wedge dA + d(A \wedge a) + dA \wedge a + 2a \wedge A \wedge A) \\ &= \frac{1}{4\pi^2} \int_M \operatorname{Tr} (a \wedge (dA + A \wedge A)) + \frac{1}{8\pi^2} \int_M d \operatorname{Tr} (A \wedge a) \\ &= \frac{1}{4\pi^2} \int_M \operatorname{Tr} (a \wedge F). \end{aligned}$$

Since a is arbitrary, we have $F = 0$.

Theorem 5.7. *The critical points of the Chern-Simons functional are exactly the flat connections on $SO(TM)$.*

5.4 Gauge Transformation

Definition 5.8. Let $\pi : P \rightarrow M$ be a G -principal bundle, the gauge transformation of P is a G -bundle automorphism $\varphi : P \rightarrow P$, i.e., φ is a diffeomorphism such that $\varphi(pg) = \varphi(p)g$ and $\pi \circ \varphi = \pi$.

$$\begin{array}{ccc} P & \xrightarrow{\varphi} & P \\ & \searrow \pi \quad \swarrow \pi & \\ & M & \end{array}$$

All gauge transformations of P form the gauge group of P , denoted by $\mathcal{G}(P)$.

Example 5.9. For trivial bundle $P = M \times G$, each smooth function $f : M \rightarrow G$ defines a gauge transformation $\varphi(x, g) = (x, f(x)g)$. In fact, all gauge transformations are of this form, so $\mathcal{G}(M \times G) \cong C^\infty(M, G)$, where multiplication in $C^\infty(M, G)$ is defined pointwise.

Let $\varphi : P \rightarrow P$ be a gauge transformation, then for any $p \in P$, there exists a unique $g_p \in G$ such that $\varphi(p) = pg_p$. Thus we can define a smooth function $f : P \rightarrow G$ by $f(p) = g_p$.

Proposition 5.10. $f : P \rightarrow G$ defined above satisfies $f(pg) = g^{-1}f(p)g, \forall p \in P, g \in G$.

Proof. Since φ is G -equivariant,

$$pgf(pg) = \varphi(pg) = \varphi(p)g = pf(p)g \implies f(pg) = g^{-1}f(p)g. \quad \square$$

The above proposition implies that f can be viewed as a section of the associated bundle $\text{Ad}P := P \times_{\text{cong}} G$ with G acting on itself by conjugation.

Theorem 5.11. *There is a one-to-one correspondence between the gauge group $\mathcal{G}(P)$ and the space of sections $C^\infty(M, \text{Ad}P)$. If we define the multiplication in $C^\infty(M, \text{Ad}P)$ pointwise, then this correspondence is a group isomorphism.*

Proof.

$$\left\{ \begin{array}{c} \text{sections of } \text{Ad}P \\ s : M \rightarrow \text{Ad}P \end{array} \right\} \xleftrightarrow{1-1} \left\{ \begin{array}{c} G\text{-equivariant maps} \\ f : P \rightarrow G \end{array} \right\} \xleftrightarrow{1-1} \left\{ \begin{array}{c} \text{gauge transformations} \\ \varphi(p) = pf(p) \end{array} \right\}. \quad \square$$

Example 5.12. For the trivial bundle $P = M \times G$, we have $\text{Ad}P = M \times G$, so $C^\infty(M, \text{Ad}P) \cong C^\infty(M, G)$.

Example 5.13. For abelian G , the conjugation action is trivial, so $\text{Ad}P = P \times_{\text{cong}} G \cong M \times G$, thus $\mathcal{G}(P) \cong C^\infty(M, G)$.

Gauge transformation acts on connections

Let $\varphi : P \rightarrow P$ be a gauge transformation, then for any connection ω on P , we can pull back ω via φ to get $\varphi^*\omega$.

Theorem 5.14. $\varphi^*\omega$ is still a connection on P .

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5.5 Holonomy and flat connections

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